

Point sur les méthodes AD/AI (MV)

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Goal of these slides

- We use the same notations used in the document "pointMethodMV.pdf". Let $\ell(\mathbf{Z}_{M,1})$ and $\ell(\mathbf{Z}_{M,2})$ two-heavy-tailed distributed random variables.
- Let $g : \mathbb{R}_+^2 \leftarrow \mathbb{R}_+$ be a function defining the effect of the radial components of a multivariate time series.
- We here try to approach the law of $\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2})$ knowing $g(\ell(\mathbf{Z}_{M,1}), \ell(\mathbf{Z}_{M,2})) > u_{\ell,g}$ when $u_{\ell,g}$ is large enough.

Introduction

- The objective is here to summarise the different case that appear (Figure 1).
- We will not use the same tools to simulate new extreme vectors.

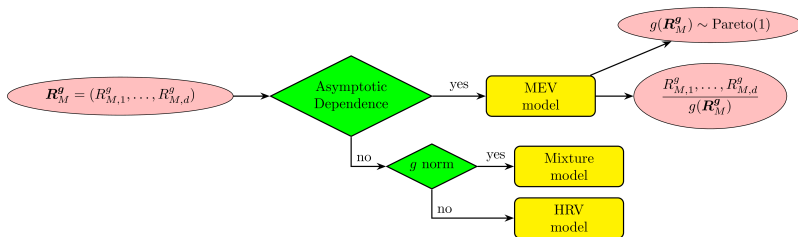


Figure 1: Summary of the different cases

Full AI

Implications of full AI case

- Let $g : \mathbb{R}_+^2 \leftarrow \mathbb{R}_+$ be a norm.
- We denote $\chi_{i,j}(u) = \mathbb{P}(X_j > u \mid X_i > u)$. $\chi_{i,j}(u) \rightarrow 0$ as $u \rightarrow +\infty$ for $i, j = 1, 2$ if X_i and X_j are asymptotically independent (AI).

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- It implies that $\mathbb{P}(\max(X) > u) = \mathbb{P}(X_1 > u) + \mathbb{P}(X_2 > u)$ for $u \rightarrow +\infty$.

Tools used

- This choice of g and the presence of asymptotic independence imply that we will use specific tools (Figure 2).
- We explain in the following slides why we can use the term "mixture".

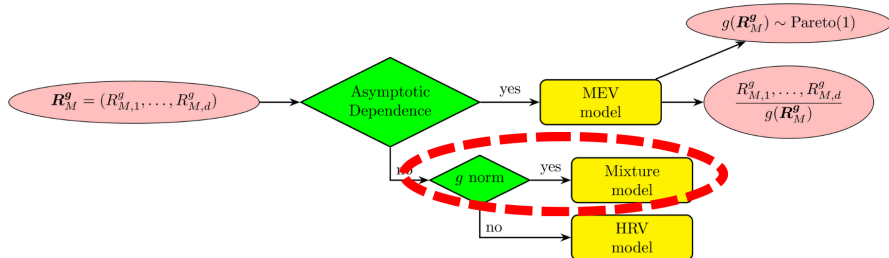


Figure 2: Focus on the AI case

Simulation of extreme events in the full AI case

- Denoting $P_1(x, y) = \mathbb{P}(X_1 \leq x, X_2 \leq y \mid X_1 > u)$ and $P_2(x, y) = \mathbb{P}(X_1 \leq x, X_2 \leq y \mid X_2 > u)$ the two conditional distributions.

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- Secondly, for $(x, y) / \max(x, y) > u$, we end up with

$$\mathbb{P}(X_1 \leq x, X_2 \leq y \mid \max(X) > u) = w_1 P_1(x, y) + w_2 P_2(x, y)$$

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- The weights w_1, w_2 represents the weights of the different scenarios.

Behaviour of each coordinate for extreme observations

- We can decompose $P_1(x, y)$ into two quantities:

$$P_1(x, y) = \mathbb{P}(X_1 \leq x \mid X_1 > u) \mathbb{P}(X_2 \leq y \mid u < X \leq x)$$

- We identify an **exceedance law** of one marginal and a **conditional law**.
- We can use this latter to describe what is the tail behaviour of the vector.

- Let show that $\mathbb{P}(X_2 \leq y \mid u < X_1 \leq x)$ will equal 1 if $(y > u)$ and $(x > u)$.

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- Since $\mathbb{P}(X_2 > y, X_1 \in (u, x]) = \mathbb{P}(X_2 > y, X_1 > u) - \mathbb{P}(X_2 > y, X_1 > x)$, we obtain

$$\mathbb{P}(X_2 > y, X_1 \in (u, x]) < \mathbb{P}(X_1 > u, X_2 > u) \rightarrow 0$$

as $u \rightarrow +\infty$.

Behaviour of the conditional law

- Let show that $\mathbb{P}(X_2 \leq y \mid u < X_1 \leq x)$ will equal 1 if $(y > u)$ and $(x > u)$.
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$$\mathbb{P}(X_2 > y, X_1 \in (u, x]) < \mathbb{P}(X_1 > u, X_2 > u) \rightarrow 0$$

as $u \rightarrow +\infty$.

- Thus, $\mathbb{P}(X_2 < y \mid u < X_1 \leq x) \rightarrow 1$ (complement of the event) when $\min(x, y) > u$ and $u \rightarrow +\infty$

- We have thus $P_1(x, y) = \mathbb{P}(X_1 \leq x \mid X_1 > u)$ for $\min(x, y) > u$.

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- It implies that when the first coordinate is above the threshold ($x > u$), the second one is strictly below it.
- Same idea for the case $X_2 > u$. **Only one** coordinate is above the threshold u for each realisation of the law.

Full and partial AD case

- $\chi_{1,2}(u) \not\rightarrow 0$ implies that $\mathbb{P}(\max(X) > u) \neq \mathbb{P}(X_2 > u) + \mathbb{P}(X_1 > u)$ when $u \rightarrow +\infty$.
- Bivariate extreme values model (bev) can be used to approach such distributions.

- It is assumed here that extreme levels of $\min(X, Y)$ can be reached (Legrand 2022).
- Using a polar coordinate representation, we simulate new angles with a parametric (Tawn 1988) or a non-parametric method (Legrand 2022).

Simulation of extreme events in the partial AD case

- In the partial AI case, knowing $X_1 > u$, only some other variables can be above u . We need to redefine what happens in a scenario.

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- Following (Mourahib, Kiriliouk, and Segers 2025), let $J_k \subset [[1, d]]$ and $C_k = \{X_j > u \ \forall j \in J_k, \ X_{j'} < u \ \forall j' \in [[1, d]] \setminus J_k\}$. C_k corresponds to an **extreme direction**.

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- We get

$$\mathbb{P}(\max(X) > u) = \sum_{k=1}^K \mathbb{P}(X \in C_k)$$

where $1 < K < d$ is the number of scenarios.

- If $K = d$, we retrieve the AI case. If $K = 1$, we retrieve the full AD structure.

Simulation of extreme events in the partial AD case

- We redefine $P_k(x_1, \dots, x_d) = \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d \mid X \in C_k)$

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- We get

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with $w_k = \mathbb{P}(X \in C_k)$ verifying $\sum_{k=1}^K w_k = 1$.

- For each $k = 1, \dots, K$, a mev model is used on the subvector X_{J_k} whereas other coordinates are below u . If we use the model of (Mourahib, Kiriliouk, and Segers 2025), these latter equal 0 (lower bound with a Frechet scale).

Heffernan model

Heffernan models for *max* based definition

- Without making assumptions on asymptotic dependence, we can rewrite the definition of extreme vectors using the formula of total probabilities:

$$\mathbb{P}(\max(X_1, X_2) > u) = \mathbb{P}(X_1 > u) + \mathbb{P}(X_1 < u, X_2 > u)$$

- We obtain from this property a new expression of the exceedances' law :

$$\mathbb{P}(X_1 \leq x, X_2 \leq y \mid \max(X_1, X_2) > u) = w_1 P_1(x, y) + w_2 P_2(x, y)$$

with $w_1 = \mathbb{P}(X_1 > u \mid \max(X_1, X_2) > u)$ and

$w_2 = \mathbb{P}(X_1 < u, X_2 > u \mid \max(X_1, X_2) > u)$

- We have also $P_1(x, y) = \mathbb{P}(X_1 \leq x, X_2 \leq y \mid X_1 > u)$ and $P_2(x, y) = \mathbb{P}(X_1 \leq x, X_2 \leq y \mid X_1 < u, X_2 > u)$

Heffernan models for max based definition

- We can use conditional model (Heffernan and Tawn 2004) to approach P_1 but we cannot do the same thing for P_2 .
- In contrast, we cannot directly use this type of model to sample from P_2 since a second condition, $X_1 < u$, does intervene.
- We can however approach the law of $(X_1, X_2 \mid X_2 > u)$ noted $\mathcal{P}$ to obtain realisations of the law of P_2 .

Simulation algorithm

- We describe here the simulation of N_{sim} extreme events.
- For $j = 1, \dots, N_{\text{sim}}$, sample C_j from a binomial law of parameter w_1 .
- If $C_j = 1$, sample from the estimation of P_1 . Otherwise, sample from $\mathcal{P}$ but keep **only** vectors verifying $X_{\text{sim},1} < u$ (**rejection sampling**).
- Return the simulated vectors $\mathbf{X}_{\text{sim}}$.

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